UTRGV Open Mathematics Competition Fall 2025

Answer format: unless otherwise stated, all answers must be simplified fully. For trigonometric functions, exact values should be given for arguments that are integer multiples of $\pi/12$; inverse trigonometric functions whose values are integer multiples of $\pi/12$ should be evaluated exactly; logarithms and exponentials should be evaluated exactly when possible; fractions should be in lowest terms; radical expressions should be in simplest form, e.g., $3\sqrt{6}$ is okay but $\sqrt{54}$ isn't; and answers should include proper units where applicable. Multiples/powers of transcendental numbers can left as is, e.g. -5π and e^2 are okay, and ratios of radicals need not be rationalized, e.g., $\frac{8}{\sqrt{23}}$ is acceptable.

(1) Calculate the mean value of the numbers: 2, 8, 11, -6, and 0.

(2) Write the number 720 as a product of powers of prime numbers, e.g., $18 = 2 \times 3^2$.

(3) If x, y, and z are real numbers where $x^2y^3z = 7$, find the value of $2x^4y^6z^2$.

(4) A right triangle has a hypotenuse length of 8 and one of its legs has a length of 4. Determine:

- the length of the other leg
- all angles in the triangle in radians

(5) A rectangle has perimeter 9 units. If the length of one side is double the length of the other, find the area of the rectangle.

(6) Find all solutions to the equation $x^2 - 2x - 1 = 0$.

(7) If $\tan\theta = 1$ and $\pi/2 < \theta < 3\pi/2$, find θ.

(8) If $f(x) = x^2$ and $g(x) = \sin x$, find $f(g(\frac{\pi}{4}))$.

(9) Order the numbers in ascending order by value: $3, \sqrt{6}, \sqrt[3]{17}, \sqrt[4]{25}, \sqrt[5]{30}$.

(10) Find real numbers a, b such that the following function is continuous and differentiable over $\mathbb{R}$

$$f(x) = \begin{cases} ax^2 + b & x \leq -1 \\ \frac{x^2-1}{x+1} & x > -1. \end{cases}$$

(11) Compute $\int t\,\mathrm{e}^{2t}\sin(2t)\,\mathrm{d}t$.

(12) Find the number of 2×2 matrices with entries equal to either 0 or 1 with determinant equal to 1.

(13) Solve $y'(x) = (x+1)y(x)^2$ with $y(0) = -1$.

(14) A company is creating ID codes of length 3 for its employees using the numbers $\{1, 2, 3, 4, 5\}$. The number 2 can appear at most once. The number 4 can appear at most once. How many ID codes are possible?

(15) Let m, j be nonnegative integers. For every m, define the following sequence

$$\begin{cases} a_m(0) = \frac{1}{2^m} \\ a_m(j+1) = a_m(j)^2 + 2a_m(j). \end{cases}$$

Find $\lim_{n\to\infty} a_n(n)$.